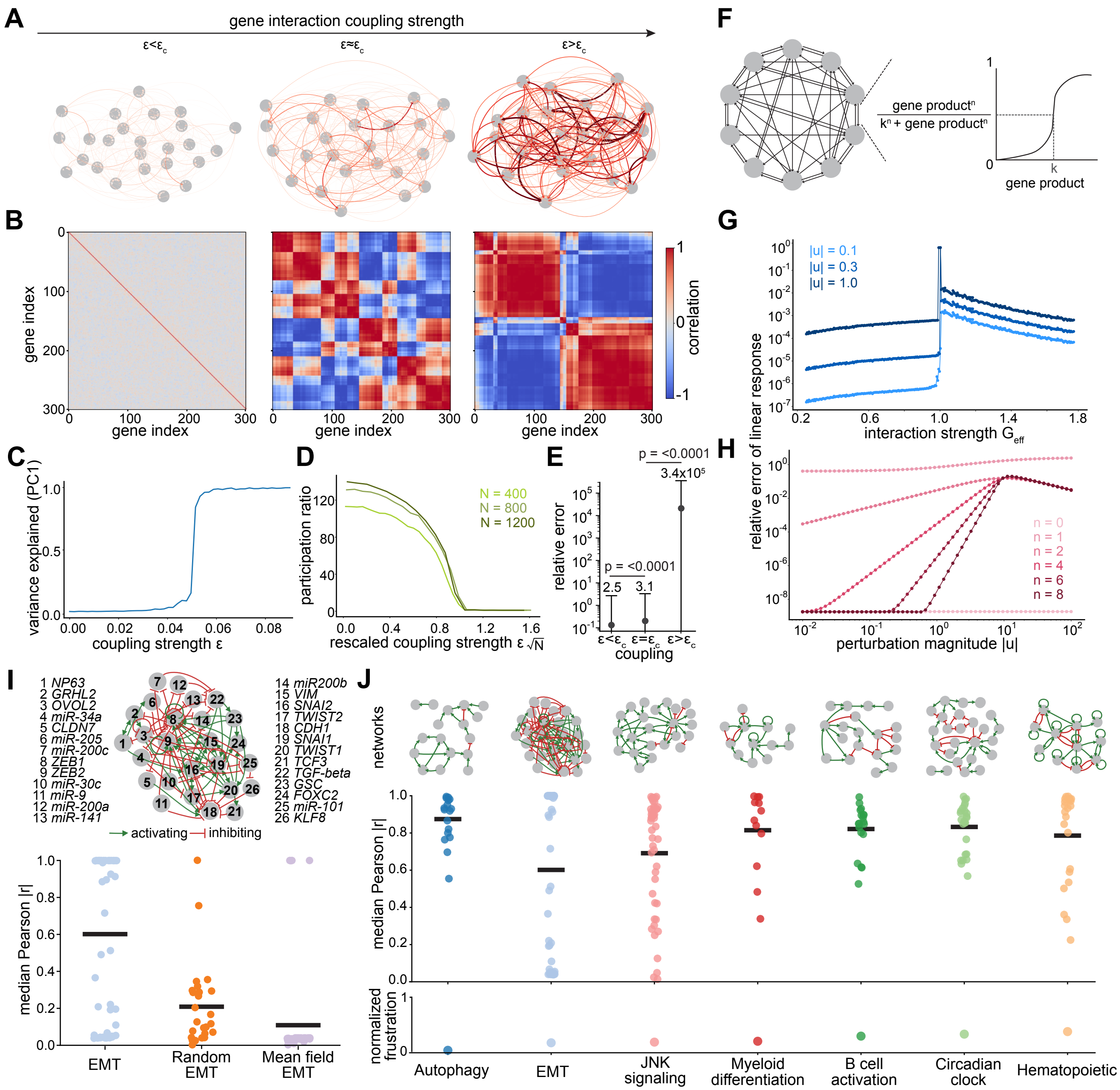

Fig. 2: Genome-wide response to perturbations in synthetic regulatory networks.

a, Random linear regulatory networks of N genes. Nodes represent genes and edge thickness represents the (absolute) strength of interactions between genes. Subcritical, critical and supercritical networks are shown from left to right. **b,** Steady-state gene-gene correlations for the networks in **a**). **c,** Fraction of variance explained by the first principal component as a function of the gene-gene interaction parameter. **d,** Participation ratio (effective dimension) as a function of the scaled interaction parameter for different sized networks (N = 400, 800, 1,200). **e,** Points are the typical error in linear response, $||\Delta X - \Sigma u||^2/||\Delta X||^2$, across all genes in the three-regimes, averaged over 500 trajectories. Standard error bars shown. (One-sided Wilcoxon signed-rank tests: subcritical error < critical error, p-value < 0.0001 ; critical error < supercritical error, p-value < 0.0001). **f,** Non-linear networks with activating Hill function interactions, n = 10. **g,** Error in linear response as a function of effective interaction strength, G_{eff} at different perturbation magnitudes (|u| = 0.1, 0.3, 1.0). Points represent individual simulation runs and lines guide the eye. **h,** Error in linear response as a function of perturbation magnitude |u| for several different Hill coefficients (n), n = 10. **i,** EMT regulatory network (top), in which genes interact by inhibitory (red) and activating (green) edges. Bottom, median absolute Pearson correlation (|r|) between predicted and observed covariance under knockdown of each gene for the EMT network, a randomized version and a mean-field covariance for the EMT network. Each point is one gene; black bars indicate the absolute Pearson correlation mean across all genes in a network. **j,** Seven biological regulatory networks including EMT (top; edges coloured as in **i**), the median absolute Pearson correlation for knockdown of each gene (middle) and the normalized mean frustration from Boolean simulations (bottom). Each point in the middle row corresponds to one gene; black bars indicate the mean across all genes in a network.